

ROAD-06-PILOT-PROGRAM-DESIGN

ROAD-06: PILOT PROGRAM DESIGN

Carrington Storm Motors — Project AEGIS

Comprehensive Pilot Program Architecture for All Major Product Lines

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Partner Liaisons: All 18 Agents of Aegis (per product line)

PILOT PROGRAM PHILOSOPHY

Pilot programs are not marketing exercises. They are rigorous scientific investigations designed to (1) validate product performance under real-world conditions, (2) generate the peer-reviewed data necessary for regulatory approval, insurance actuarial modeling, and customer trust, (3) identify failure modes that cannot be detected in laboratory testing, and (4) establish the reference installations that will anchor regional market expansion. Every CSM pilot follows a standardized architecture: 6-month preparation phase, 12-month active data collection, 3-month analysis and publication, with pre-defined success criteria, statistical analysis plans, and independent scientific oversight.

PILOT 1: AEGIS HOME — RESIDENTIAL PILOT PROGRAM

P1.1 Program Overview

Scope: Construction and monitoring of 50 Aegis Homes across three geographic regions
Timeline: 6-month prep (Q3 2027–Q4 2027), 12-month data collection (Q1 2028–Q4 2028), 3-month analysis (Q1 2029)
Total Duration: 21 months
Budget: \$6.2M
Lead Agent: Operative Corvus Nyx (AEGIS HOME)
Scientific Oversight: Independent research university (target: University of Washington — Cascadia Subduction Zone expertise; Georgia Tech — power systems; or UC San Diego — structural engineering)

P1.2 Site Selection

Region	Number of Homes	Rationale	GIC Risk Profile	Site Partners
Pacific Northwest (WA/OR)	20 homes	Cascadia Subduction Zone — highest GIC risk in continental US; strong environmental consciousness; green building mandate alignment	HIGH — coastal proximity, igneous rock geology (high resistivity, channels GIC into power grid)	D.R. Horton (production builder), local electric utilities (Puget Sound Energy, Pacific Power)
Upper Midwest (MN/WI)	15 homes	High GIC risk region (Precambrian shield geology); severe winter climate for thermal performance testing; strong utility engagement	HIGH — igneous/metamorphic rock, long EHV transmission lines	Pulte Homes, Xcel Energy, Dairyland Power Cooperative

Gulf Coast (TX/LA)	15 homes	Hurricane + EMP dual threat market; high humidity for MXene durability testing; diverse demographic for equity analysis	MODERATE — sedimentary basin (lower GIC but increasing with grid expansion)	Lennar, Entergy, CenterPoint Energy
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Site Selection Criteria: 1. Within 5 km of geomagnetic observatory or utility-owned GIC monitor for reference data 2. Diverse housing types: 30 single-family detached, 10 townhouse/duplex, 10 multifamily (apartment retrofit) 3. Representative of US Census income distribution (30% low-income, 40% middle-income, 30% upper-income) 4. Accessible for public tours and media (community acceptance requirement) 5. Signed homeowner agreement for 12-month monitoring with privacy protections

P1.3 Data Collection Plan

Data Category	Sensors per Home	Measurement	Frequency	Method
Ground-Induced Current	4 (N-S, E-W, vertical, reference)	Current density (A/m ²), electric field (mV/km)	1 Hz continuous	Magnetotelluric sensors + utility reference data
Panel Shielding Effectiveness	2 (wall interior vs. exterior field probe)	SE in dB, 1 kHz-10 GHz spot checks quarterly	Monthly	Handheld spectrum analyzer + reference antenna
Indoor Environmental Quality	6 (temperature, humidity, CO ₂ , VOCs, PM2.5, radon)	Per EPA Indoor airPLUS standards	5-minute intervals	Commercial IAQ monitors (Awair, Airthings)
Energy Usage	Whole-home + 6 circuit-level	kWh	15-minute intervals	Utility smart meter + Sense energy monitor
Structural Health	12 fiber optic strain gauges (foundation + walls)	Microstrain	1 Hz continuous	FBG interrogator (Luna Innovations)
Water Quality	2 (incoming + outgoing from GFRP pipe system)	pH, conductivity, chlorine residual, heavy metals	Weekly grab sample + continuous conductivity	In-line sensors + lab analysis
Occupant Comfort / Satisfaction	Monthly survey + continuous wearable (optional)	Thermal comfort (ASHRAE 55), acoustic, visual, overall satisfaction	Monthly	Validated post-occupancy evaluation (POE) survey instruments
Health Outcomes	Optional: heart rate variability (HRV), sleep quality, stress biomarkers	RMSSD, sleep duration/efficiency, cortisol (hair sample)	Pre-installation baseline + quarterly	Wearable (Oura/Whoop) + lab analysis
Maintenance Events	Logging by homeowner	Date, component, description, resolution	Continuous	Mobile app (Marmalade maintenance module)
Cost Data	Utility bills, maintenance costs, insurance premiums	\$	Monthly	Utility portal API + homeowner self-report

Data Volume Estimate: ~2.5 TB per home per year (primarily from fiber optic strain gauge and IAQ sensor streams). Total study data: ~125 TB.

P1.4 Success Criteria

Criterion	Threshold	Measurement Method	Statistical Analysis
Primary: GIC Reduction	>94% reduction in induced current vs. control home (conventional construction, matched location)	Magnetotelluric sensor comparison between Aegis Home and control home at same substation feeder	Paired t-test, $\alpha=0.05$, $\beta=0.20$ (80% power), minimum detectable effect = 5% difference

Secondary: No Electrical Safety Events	Zero shock incidents, zero fire events attributable to the dielectric electrical system	Incident reporting + annual NEC inspection by licensed electrician	Descriptive — any event triggers root cause analysis and potential design change
Secondary: Energy Performance	HERS Index <45 (vs. US average 100)	RESNET HERS rater assessment	One-sample t-test against HERS 100 baseline
Secondary: Indoor Air Quality	EPA Indoor airPLUS certification achieved	Independent verification	Pass/fail per EPA criteria
Secondary: Occupant Satisfaction	>90% of occupants “satisfied” or “very satisfied” at 12 months	Validated POE survey (7-point Likert scale)	One-proportion z-test against 80% benchmark
Secondary: Durability	<2% of monitored components exhibiting degradation requiring repair/replacement at 12 months	Maintenance log analysis	Descriptive with 95% confidence interval
Secondary: Insurance Premium Impact	Documented premium reduction of >5% for Aegis Home vs. comparable conventional home	Insurance carrier premium data (anonymized)	Independent samples t-test

P1.5 Control Group Design

For each Aegis Home, a matched control home will be identified: - Same geographic area (within 5 km, same substation feeder) - Same construction type (single-family, townhouse, apartment) - Same approximate age (± 5 years), square footage ($\pm 15\%$), and number of occupants - Conventional construction (no EM hardening) - Control homes receive same sensor package, with homeowner compensation (\$500/year + free energy audit) - 25 control homes (matched to instrumented Aegis Homes; 25 Aegis Homes matched to control; remaining 25 Aegis Homes unmatched — used for internal variability analysis)

PILOT 2: INFRASTRUCTURE HARDENING PILOT

P2.1 Program Overview

Scope: 5 electrical substations, 3 highway/railway bridges, and 2 pipeline sections
Timeline: 6-month prep (Q2 2027–Q4 2027), 12-month monitoring (Q1 2028–Q4 2028), 6-month analysis (Q1–Q2 2029)
Total Duration: 24 months
Budget: \$8.5M
Lead Agent: Engineer Orion Cross (SILENCE) — infrastructure engineering
Utility Partners: To be identified (target 3 electric utilities, 2 state DOTs, 1 gas pipeline operator)

P2.2 Infrastructure Pilot Sites

Site Type	Number	Product Deployed	Location Rationale	Partner
Electrical Substation	5	Aegis Fencing (BFRP posts + MXene mesh canopy), Transformer dielectric isolation pads (Si ₃ N ₄ /PTFE), Bus bar insulation upgrade	GIC-prone regions (2 in Pacific NW, 2 in Upper Midwest, 1 in Northeast — Quebec border for Hydro-Québec learnings)	Electric utilities (TBD)
Highway Bridge	2	Bridge bearing dielectric pads (AASHTO compliant), BFRP-reinforced deck sections, Geopolymer concrete pier jackets	Seismically active + GIC-prone (Cascadia and New Madrid zones)	State DOTs (WSDOT, MoDOT)
		Railway signal		

Railway Bridge	1	protection (PMMA POF signal cables), Track circuit isolation, BFRP bridge ties	Northeast Corridor (Amtrak) — highest consequence of signal failure	Amtrak / FRA
Natural Gas Pipeline	2 (10 km each)	GFRP dielectric joints at 500 m intervals, Cathodic protection compatibility verification	Pipeline in GIC-prone region with cathodic protection system (required for GFRP/pipeline interface study)	Pipeline operator (Kinder Morgan, Enbridge, or similar)

P2.3 Infrastructure Success Criteria

Criterion	Threshold	Method
Substation GIC Reduction	>90% reduction in transformer neutral current vs. pre-installation baseline	Utility SCADA data + dedicated GIC monitors (Neutral current transformers)
Bridge Bearing Dielectric	>50 kV/mm dielectric withstand sustained for 12 months; no measurable GIC current through bearing	In-situ hipot testing + embedded current sensors
Railway Signal Reliability	Zero signal anomalies attributable to GIC during 12-month monitoring	Signal event recorder data + GIC correlation analysis
Pipeline Cathodic Protection	CP system effectiveness maintained at NACE SP0169 criteria (off-potential –850 mV vs. Cu/CuSO ₄) with GFRP joints installed	Close-interval survey pre- and post-installation

PILOT 3: CHARLEMAGNE FLEET PILOT

P3.1 Program Overview

Scope: Complete electromagnetic hardening retrofit of ONE Charlemagne-class cruise vessel (100,000 GT class, ~3,000 passenger capacity)

Timeline: 6-month prep (engineering + materials procurement; Q3 2027–Q4 2027), 18-month retrofit (Q1 2028–Q2 2029), 12-month operational monitoring (Q3 2029–Q2 2030)

Total Duration: 36 months

Budget: \$47M (vessel retrofit) + \$3.8M (monitoring and analysis) = \$50.8M

Lead Agent: Captain Meridia Vaun (CHARLEMAGNE)

Cruise Line Partner: To be identified (target: Carnival Corporation [brands: Princess, Holland America, Carnival], Royal Caribbean Group, or Norwegian Cruise Line Holdings — all three have Alaska/Pacific NW itineraries in GIC-prone corridors)

P3.2 Vessel Selection Criteria

- Vessel scheduled for dry dock during Q1 2028–Q2 2029 (routine maintenance window — retrofit piggybacks on planned dry dock)
- Vessel operating in GIC-prone latitudes (>45°N or high-resistivity seafloor regions)
- Vessel with cooperative engineering team and committed executive sponsor
- Insurance carrier for vessel willing to provide premium data for before/after comparison
- Classification society (ABS, DNV, or Lloyd’s) willing to engage in the new technology qualification process

P3.3 Retrofit Scope per Vessel

System	Component	Material	Installation Scope	Verification Method
Hull	MXene spray coating (92 dB SE at 45 μm)	Ti ₃ C ₂ T _x MXene, epoxy binder	Full underwater hull + topsides, ~25,000 m ²	Post-coating SE measurement via portable test fixture (every 100 m ²)

Structure	BFRP stiffener replacement at identified conductive paths	BFRP/Elum pultruded profiles	Replace all steel-to-steel conductive paths at 200+ identified locations	DC resistance measurement ($>10^8 \Omega$ target)
Wiring	PMMA POF replacement for all communication cables	PMMA core, MXene-coated, 62 fibers/harness	47.3 km of copper communication cable replaced	Optical time-domain reflectometer (OTDR) verification per cable
Power	LiFePO ₄ battery architecture for hotel loads	LiFePO ₄ cells, ZrO ₂ enclosure, GIC-hardened BMS	20 MWh battery installation in former ballast tank compartment	Full charge/discharge cycling, BMS fault injection testing
Propulsion (emergency)	Piezoelectric emergency propulsion	KNbO ₃ -BaTiO ₃ motor (2,000 HP target — note: TRL 2, may not be ready for pilot; fallback = hybrid diesel-electric with dielectric isolation)	Emergency propulsion pod, deployable	Sea trial under simulated GIC
Navigation	Fiber optic gyroscope (FOG) + PMMA POF data network	Commercial FOG (KVH, Northrop Grumman) with GIC-hardened power supply	Bridge navigation suite	Dockside + sea trial accuracy verification
Safety	Dielectric fire suppression pumps, Protonic emergency communication	Ceramic impeller pumps, H-FET transceivers	5 fire zones, 12 communication nodes	System operational test
Entertainment	PMMA POF backbone for passenger Wi-Fi, in-cabin entertainment	POF with MXene EMI shielding	3,000+ cabins, 20 public spaces	Bandwidth + uptime testing

P3.4 Charlemagne Pilot Success Criteria

Criterion	Threshold	Method
Hull-Induced Current	$<0.03\%$ of unprotected baseline (measured via hull voltage differential during geomagnetic storm)	Hull voltage sensors (bow-to-stern, port-to-starboard) correlated with geomagnetic observatory data
Passenger Safety	Zero electromagnetic safety incidents during 12 months of commercial operation	Incident reporting + passenger/crew survey
Insurance Premium Impact	$>12\%$ reduction in hull and P&I (Protection & Indemnity) insurance premium for hardened vessel	Before/after premium comparison from vessel's insurance carrier
Classification Society Approval	ABS/DNV/Lloyd's class certificate issued with EM hardening notation	Formal class certificate
Operational Availability	$<1\%$ additional downtime attributable to EM hardening systems (i.e., systems do not reduce vessel operational availability)	Voyage data recorder analysis
Crew/Passenger Acceptance	$>85\%$ crew and passenger satisfaction with no negative impact on cruise experience	Survey instrument

PILOT 4: ROBOTICS PILOT — PHANTOM MK-1

P4.1 Program Overview

Scope: Deployment of 10 Phantom MK-1 robots in electric utility environments for post-storm inspection and maintenance
Timeline: 3-month prep (Q3 2028–Q4 2028), 6-month field deployment (Q1 2029–Q2 2029), 3-month analysis (Q3 2029)
Total Duration: 12 months
Budget: \$8.5M (10 Phantom units at \$180K each = \$1.8M; telemetry infrastructure \$2.2M; utility partner coordination \$1.5M; field support team \$2.0M; data analysis and reporting \$1.0M)
Lead Agent: Agent Nyx Calder (PHANTOM)
Utility Partners: 3 electric utilities in GIC-prone regions (Pacific NW, Upper Midwest, Northeast)

P4.2 Phantom Pilot Deployment Scenarios

Scenario	Number of Deployments	Environment	Success Criteria
Post-Storm Damage Assessment	20 (across 3 utilities, ~7 per utility)	Downed power lines, flooded substations, debris fields	Autonomous navigation in EMI + physical damage environment; identification of 95% of damage conditions vs. human inspector
Substation Inspection (Routine)	30 (monthly deployments at 3 substations per utility)	Operating substation, 60 Hz fields, partial discharge RF noise	Thermal imaging anomaly detection; >90% match with human inspector findings
Transmission Line Right-of-Way Survey	15	Forested, mountainous terrain, variable weather	Autonomous path planning; vegetation encroachment detection
Emergency Communication Relay	10 simulated events	Wide-area EMI, GPS-denied	Maintain PMMA POF or protonic mesh communication relay; zero dropped data packets
Combined-Environment Stress Test	5 simulated events	Simultaneous EMI + rain + night operations	Full mission completion without degraded-mode fallback

P4.3 Phantom Pilot Data Collection

Data Category	Sensors	Data Volume per Robot per Day
Actuator Performance (32 joints)	Torque, position, temperature, current draw	500 MB
EMI Environment	Onboard GIC sensors, spectrum analyzer	200 MB
Camera/Imaging	Thermal (FLIR), visual (4K), LIDAR point cloud	15 GB (raw), 2 GB (compressed)
Navigation	GPS, IMU, wheel odometry, SLAM map	300 MB
Communication	POF data bus traffic, protonic radio link quality	50 MB
Battery / Power	State of charge, temperature, cell voltages	10 MB
TOTAL per robot per day		~18 GB raw / ~3 GB processed

Total Pilot Data Volume: ~5.4 TB for 10 robots over 6 months.

PILOT 5: PROTONIC COMMUNICATIONS PILOT

P5.1 Program Overview

Scope: Deployment of 5 protonic communication nodes in field environment

Timeline: 6-month prep (Q3 2027–Q4 2027), 12-month operation (Q1 2028–Q4 2028), 3-month analysis (Q1 2029)
Total Duration: 21 months
Budget: \$3.8M
Lead Agent: Agent Solara Vance (PROTON)

P5.2 Node Deployment Configuration

Node	Location	Deployment Type	Coverage Target
Node 1 (Master)	CSM El Segundo facility	Rooftop, grid-connected for baseline, piezo backup	Reference node — continuous calibration
Node 2	Utility substation (Pacific NW)	Co-located with Infrastructure Pilot substation	GIC monitoring communication relay
Node 3	Rural community center (Cascadia region)	Standalone, 100% piezo-powered, no grid connection	Post-disaster community communication hub
Node 4	Charlemagne Pilot Vessel	Shipboard, integrated with vessel Protonic network	Maritime mobile node — communication during GIC events at sea
Node 5	Emergency operations center (FEMA Region X)	Co-located with EOC backup communications	Government emergency communication integration

P5.3 Protonic Pilot Success Criteria

Criterion	Threshold	Method
Uptime	>99.5% over 12 months (excluding planned maintenance)	Automated ping/heartbeat monitoring
Voice Quality	MOS (Mean Opinion Score) >3.5 at 3.4 kHz bandwidth	Subjective listening tests with trained evaluators quarterly
Text Reliability	Bit error rate <10 ⁻⁶ at 50 baud	Automated BER tester over 10 ⁶ bits per test
Range	>5 km node-to-node	RSSI measurement at graduated distances
Piezo Power Sustainability	Zero battery replacement or external charging events over 12 months	Power system telemetry
Interoperability	Successful voice bridge between Protonic Network and conventional VHF radio (via gateway)	Quarterly interoperability drill
FCC Compliance	No regulatory enforcement actions; FCC petition progressed	FCC correspondence tracking

PILOT 6: AEGIS EMBARK — VEHICLE PILOT

P6.1 Program Overview

Scope: Installation of all 8 Aegis Embark products in 50 fleet vehicles (25 electric, 25 internal combustion) and 25 consumer vehicles
Timeline: 6-month prep (Q4 2027–Q1 2028), 12-month monitoring (Q2 2028–Q1 2029), 3-month analysis (Q2 2029)
Total Duration: 21 months
Budget: \$2.8M
Lead Agents: All 8 Aegis Embark Agents (Nash, Kade, Draven, Cross, Arden, Voss, Fen, Dorne) — each responsible for their product’s pilot segment

P6.2 Fleet Partners

Fleet Type	Number of Vehicles	Vehicle Type	Partner Type
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Electric Utility Fleet	15	Ford F-150 Lightning / Chevrolet Silverado EV	Utility partner fleet vehicles (field service trucks)
Rideshare Fleet	15	Tesla Model 3/Y, Toyota Camry Hybrid	Rideshare driver partners (Uber/Lyft)
Long-Haul Trucking	10	Freightliner eCascadia, Volvo VNR Electric	Commercial fleet operator
Consumer (Early Adopter)	25	Various EV + ICE vehicles	Consumer volunteers recruited via CSM marketing channels
Government Fleet	10	Various (city/county fleet vehicles)	Municipal fleet manager partnership

P6.3 Aegis Embark Success Criteria

Product	Key Metric	Threshold	Agent
Neural-Grip MRE Steering Wrap	Vibration reduction at 125 Hz	>90% reduction vs. baseline (pre-installation)	Nash
Cervical-Guard Headrest	Driver heart rate variability (RMSSD)	>15% improvement vs. baseline during 2-hour drive	Kade
Mag-Float Seat Rail	Seat adjustment force	<5 N (vs. 15–25 N for mechanical rails)	Draven
Silence-Block LRAM Door Insert	Cabin noise at 80–100 Hz	>15 dB reduction vs. baseline	Cross
Thorax-Calm STF Seatbelt	Belt webbing load during 15 mph crash test (sled)	>50% increase in energy absorption vs. standard belt	Arden
Stellar-Tint EV Window Film	Cabin solar heat gain	>40% reduction vs. clear glass; >80% NIR rejection	Voss
Fluid-Damp Gear Shift	Shift feel quality (subjective rating)	>4.0/5.0 average driver rating	Fen
Ulnar-Rest CLD Armrest	Vibration at armrest surface (10–50 Hz)	>80% reduction vs. baseline	Dorne

PILOT PROGRAM BUDGET SUMMARY

Pilot Program	Prep	Active Phase	Analysis	Total Budget	Primary Funding Source
P1: Aegis Home (50 homes)	\$1.8M	\$3.5M	\$900K	\$6.2M	Series A + utility partner co-funding
P2: Infrastructure (5 substations, 3 bridges, 2 pipelines)	\$2.2M	\$5.1M	\$1.2M	\$8.5M	Series A + DOE/state grants
P3: Charlemagne Fleet (1 vessel)	\$3.5M retrofit prep + \$47M vessel = \$50.5M	\$3.5M monitoring (included)	\$800K	\$50.8M	Series B/C + cruise line co-investment
P4: Phantom Robot (10 units)	\$2.5M	\$4.8M	\$1.2M	\$8.5M	Series A + utility partner co-funding
P5: Protonic Communications (5 nodes)	\$1.2M	\$2.1M	\$500K	\$3.8M	Series A + NSF/FCC grants
P6: Aegis Embark	\$800K	\$1.6M	\$400K	\$2.8M	Series A + fleet

(100 vehicles)	\$500K	\$1.0M	\$1.0M	\$2.0M	partner co-funding
TOTAL PILOT PROGRAM				\$80.6M	

Total Pilot Program: \$80.6M over 21-36 months (staggered start dates)

PILOT PROGRAM — DATA MANAGEMENT AND PUBLICATION PLAN

Data Management

1. All pilot data stored in CSM Digital Twin platform (cloud-hosted, encrypted)
2. Raw sensor data retained for 10 years minimum
3. Data accessible to independent researchers via Data Use Agreement (non-commercial research only)
4. Personally identifiable information (PII) from homeowner/occupant data never shared; aggregate statistics only in publications

Publication Plan

Publication	Target Journal / Venue	Data Source	Timeline
“Measured GIC Reduction in Aegis Home Residential Buildings: 50-Home Controlled Pilot Study”	IEEE Transactions on Electromagnetic Compatibility or Building and Environment	P1: Aegis Home data	Q2 2029
“Carrington-Hardened Cruise Vessel: First-Year Operational Data from Charlemagne Pilot”	Marine Structures or Ocean Engineering	P3: Charlemagne data	Q3 2030
“Field Validation of a GIC-Immune Humanoid Robot for Post-Storm Utility Inspection”	IEEE Robotics and Automation Letters	P4: Phantom data	Q4 2029
“Protonic Communication Network: 12-Month Field Trial of a Non-Electromagnetic Disaster Communication System”	Nature Communications or IEEE Communications Magazine	P5: Protonic data	Q2 2029
“Human Factors of Carrington-Hardened Vehicle Interiors: Driver Physiological Response to Aegis Embark Products”	Ergonomics or Applied Ergonomics	P6: Aegis Embark data	Q3 2029
“Geomagnetically Induced Current Reduction via Dielectric Infrastructure Retrofits: Substation, Bridge, and Pipeline Results”	IEEE Transactions on Power Delivery	P2: Infrastructure data	Q3 2029

Independent Scientific Advisory Board

To ensure credibility of pilot results, CSM will establish an independent Scientific Advisory Board of 5-7 researchers in geomagnetism, power systems, building science, and statistics. Board members will:

- Review and approve study designs before pilot initiation
- Have unrestricted access to raw data (under confidentiality agreement)
- Be co-authors on all publications
- Receive honoraria (\$15K/year) and travel support
- Have no financial interest in CSM (no equity, no consulting fees beyond honoraria)

PILOT PROGRAM — RISK REGISTER

Risk	Probability	Impact	Mitigation
Utility partner withdrawal	MEDIUM (25%)	Loss of pilot site; schedule delay	Over-recruit: 5 utilities for 3 slots; contingency sites identified
Homeowner dissatisfaction / withdrawal	MEDIUM (20%)	Data gap; negative publicity	Generous compensation (\$500/year + free energy audit); responsive maintenance hotline; clear exit option (equipment removal at CSM cost)
Pilot product performance below specification	MEDIUM (30%)	Failure to meet success criteria; investor confidence	Pre-pilot laboratory testing gate (TRL 6 minimum before pilot); real-time monitoring to detect issues early; design iteration during pilot (acceptable if documented)
Cruise line partner not secured for Charlemagne pilot	HIGH (40%)	Schedule delay for entire Fleet program	Engage 3 cruise lines in parallel; consider government vessel (US Navy, USCG, NOAA) as alternative pilot platform
Data quality issues (sensor failure, data gaps)	MEDIUM (25%)	Reduced statistical power; publication delays	Redundant sensors on critical measurements; weekly data quality audits; 20% over-instrumentation budget
Regulatory / permitting delays for construction	MEDIUM (30%)	Schedule slip	Early engagement with building departments; pre-permitted prototype design; use of existing home retrofit (not new construction) for Phase 1 to reduce permitting complexity

“A pilot program is a promise made in data. Every sensor, every homeowner consent form, every peer-reviewed publication is a brick in the wall of credibility that separates a real company from a science project. When an insurance actuary asks for evidence that Aegis Home reduces GIC by 94%, we will not hand them a simulation. We will hand them 125 terabytes of peer-reviewed, independently verified field data from 50 homes over 12 months. That is the difference between a product and a proof of concept.” — Specialist Aura Kind (EMBARK)

Document generated 2026-07-15. All pilot budgets are ±25% at P50 confidence. Partner identification is preliminary — formal MOUs required before pilot initiation. Statistical analysis plans to be finalized with independent biostatistician/epidemiologist review prior to data collection start. All human subjects research components (homeowner surveys, health outcome monitoring) require IRB approval from partner university.